

Customer Behavior Analysis

Customer Shopping Behavior & Revenue Insights

Project Type	Customer behavior analytics and interactive dashboard
Dataset	3,900 customer purchase records
Primary Tools	PostgreSQL / SQL, Python (Pandas), Power BI
Author	Anurag Nagarkoti
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1. Executive Summary

This project analyzes 3,900 customer purchase records to understand customer behavior and revenue patterns. The analysis combines PostgreSQL/SQL business questions with Python-based data preparation and a visual dashboard for decision-oriented reporting.

The dataset contains 3,900 records across 18 original columns. The average purchase amount is \$59.76, the average review rating is 3.75/5, and total recorded revenue from purchase amounts is \$233,081.

Key headline: 27% of customers are subscribers, while the average purchase amount of subscribers (\$59.49) is slightly below that of non-subscribers (\$59.87).

2. Project Objectives

- Measure overall customer, purchase, revenue, and review performance.
- Understand subscription behavior and compare subscribers with non-subscribers.
- Identify the categories and age groups contributing the most revenue.
- Analyze purchasing patterns by gender, shipping type, discounts, and previous purchases.
- Create business-ready metrics and an interactive dashboard for quick exploration.
- Translate SQL and Python analysis into clear, decision-oriented insights.

3. Dataset Overview

The source data is a customer shopping behavior dataset. Each row represents a customer purchase record, with attributes covering demographics, products, transaction value, service choices, promotional behavior, and purchasing frequency.

Metric	Value
Rows / purchase records	3,900
Original columns	18
Age range	18–70 years
Average age	44.07 years
Average previous purchases	25.35
Categories	4
Products / items	25
Locations / states	50
Shipping types	6
Payment methods	6

4. Data Preparation & Cleaning

The Python notebook loads the CSV into Pandas, inspects the structure, checks data types and descriptive statistics, identifies missing values, and then prepares the dataset for analysis.

- The dataset contains 3,900 records and 18 original columns.
- Review Rating is the only field with missing values: 37 records are missing ratings.
- Missing review ratings are imputed using the median review rating within each product category.
- Column names are standardized to lowercase with underscores, and Purchase Amount (USD) is renamed to purchase_amount.
- Promo Code Used is removed after checking that it matches Discount Applied for all records.

5. Feature Engineering

Two analytical fields are created in the notebook:

Feature	Method / Purpose
age_group	Quartile-based segmentation into Young Adult, Adult, Middle Aged, and Senior.
purchase_frequency_days	Maps purchase-frequency labels to approximate day intervals: 7, 14, 30, 90, and 365 days.

6. Analytical Dataset

After preparation, the working dataset contains the original business attributes plus age_group and purchase_frequency_days. The analysis therefore supports both categorical comparisons and numeric summaries.

Field	Field	Field
customer_id	age	gender
item_purchased	category	purchase_amount
location	size	color
season	review_rating	subscription_status
shipping_type	discount_applied	previous_purchases
payment_method	frequency_of_purchases	age_group
purchase_frequency_days		

7. SQL Analysis & Business Questions

The PostgreSQL SQL file contains ten business questions covering revenue by gender, discount behavior, product ratings, shipping comparisons, subscription performance, customer segmentation, category-level product rankings, repeat buyers, and revenue by age group.

#	Business Question
1	Revenue generated by male vs. female customers
2	Customers using discounts while spending above average
3	Top 5 products by average review rating
4	Average purchase amount: Standard vs. Express shipping
5	Subscriber vs. non-subscriber spending and revenue
6	Products with the highest discount-purchase percentage
7	New, Returning, and Loyal customer segmentation
8	Top 3 purchased products within each category
9	Repeat buyers (>5 previous purchases) and subscription status
10	Revenue contribution by age group

8. Core Business Metrics

KPI	Result
Total customers / records	3,900
Total recorded revenue	\$233,081
Average purchase amount	\$59.76
Average review rating	3.75 / 5
Subscribers	1,053 (27%)
Non-subscribers	2,847 (73%)
Discount-applied purchases	43%

9. Revenue & Category Analysis

Category	Records	Revenue	Avg. Purchase
Clothing	1,737	\$104,264	\$60.03
Accessories	1,240	\$74,200	\$59.84
Footwear	599	\$36,093	\$60.26
Outerwear	324	\$18,524	\$57.17

Clothing is the largest category by both record volume (1,737 records) and revenue (\$104,264). Accessories is the second-largest revenue contributor at \$74,200. Footwear has a slightly higher average purchase amount (\$60.26) than Clothing, but its overall revenue is lower because it has fewer records.

10. Dashboard Walkthrough

The dashboard provides an interactive view of customer purchasing, subscription, and revenue behavior. The visible controls allow users to filter by Subscription Status, Gender, Category, and Shipping Type.



Dashboard preview supplied with the project files.

Dashboard Element	Purpose
Number of Customers	Shows customer-record volume for the current filter context.
Average Purchase Amount	Tracks the typical transaction value.
Average Review Rating	Summarizes customer feedback quality.
Subscription Status	Shows the subscriber / non-subscriber mix.
Sales by Category	Compares purchase-record volume across categories.
Revenue by Age Group	Highlights revenue contribution across age segments.
Revenue by Category	Shows category-level revenue contribution.
Sales by Age Group	Compares transaction volume across age segments.

11. Key Analytical Findings

11.1 Subscription Behavior

Subscribers account for 27% of the dataset (1,053 customers), while non-subscribers account for 73% (2,847). Average spend is very similar: \$59.49 for subscribers versus \$59.87 for non-subscribers. In this dataset, subscription status is therefore not associated with a meaningful increase in average purchase value.

11.2 Gender & Revenue

Gender	Records	Revenue	Avg. Purchase
Female	1,248	\$75,191	\$60.25
Male	2,652	\$157,890	\$59.54

Male customers generate the larger absolute revenue total (\$157,890) because they represent more records. Female customers have the slightly higher average purchase amount (\$60.25 vs. \$59.54), so volume and customer-level value should be interpreted separately.

11.3 Age Group Revenue

Age Group	Records	Revenue	Avg. Purchase
Young Adult	1,028	\$62,143	\$60.45
Adult	942	\$55,978	\$59.42
Middle Aged	986	\$59,197	\$60.04
Senior	944	\$55,763	\$59.07

Young Adults contribute the highest revenue at \$62,143, followed by Middle Aged customers at \$59,197. The average purchase values across age groups are relatively close, indicating that revenue differences are driven more by record volume than by large differences in average transaction value.

11.4 Shipping Behavior

Shipping Type	Records	Avg. Purchase
2-Day Shipping	627	\$60.73
Express	646	\$60.48
Free Shipping	675	\$60.41
Store Pickup	650	\$59.89
Next Day Air	648	\$58.63
Standard	654	\$58.46

2-Day Shipping has the highest average purchase amount (\$60.73), while Standard has the lowest (\$58.46). The difference is modest, so shipping type should be treated as a behavioral dimension rather than a proven driver of higher spend.

12. Customer Loyalty & Repeat Purchasing

Customer Segment	Customers
New	83
Returning	701
Loyal	3,116

The SQL segmentation defines New customers as having exactly 1 previous purchase, Returning customers as having 2–10 previous purchases, and Loyal customers as having more than 10 previous purchases.

Using that definition, 3,116 customers are Loyal, 701 are Returning, and 83 are New. This indicates that the dataset is dominated by customers with a substantial purchase history.

13. Review & Product Insights

Rank	Product	Avg. Review Rating
1	Gloves	3.86 / 5
2	Sandals	3.84 / 5
3	Boots	3.82 / 5
4	Hat	3.80 / 5
5	Skirt	3.79 / 5

The highest average product ratings are associated with Gloves, Sandals, Boots, Hat, and Skirt. These rankings can support product-level merchandising, promotion, and inventory discussions, while recognizing that rating differences are relatively small.

14. Business Impact & Recommendations

- Use category-level revenue and volume together when prioritizing inventory and merchandising decisions.
- Treat subscription as a retention/relationship metric rather than assuming it automatically increases transaction value.
- Use age-group and gender views for targeted campaigns, but distinguish customer volume from average spend.
- Evaluate shipping options with both purchase value and operational cost before changing delivery strategy.
- Use repeat-purchase segmentation to design differentiated offers for New, Returning, and Loyal customers.
- Monitor highly rated products as potential candidates for stronger merchandising and cross-selling.

15. Limitations

- The dataset is a transactional customer-shopping sample rather than a full longitudinal customer history.
- Revenue is calculated as the sum of Purchase Amount (USD) and does not include costs, margins, returns, or discounts as monetary adjustments.
- Age groups are created using quartiles, so their boundaries are data-driven rather than fixed business age bands.
- The SQL analysis identifies relationships and patterns; it does not establish causation.
- The dashboard is intended for descriptive and exploratory analysis, not predictive modeling.

16. Skills Demonstrated

Skill Area	How It Was Used in the Project
SQL / PostgreSQL	Business queries using GROUP BY, CASE, subqueries, window functions, ranking, and aggregation.
Python / Pandas	Data loading, inspection, missing-value handling, transformation, feature engineering, and aggregation.
Data Cleaning	Missing-value treatment, column standardization, and duplicate/redundant-field validation.
Business Analysis	Revenue, subscription, product, demographic, shipping, discount, and loyalty analysis.
Data Visualization	Interactive dashboard using KPIs, slicers, donut charts, category comparisons, and age-group analysis.
Communication	Converted analytical outputs into clear business-focused insights and recommendations.

17. Project Outcome

The project establishes an end-to-end customer behavior analytics workflow: raw customer shopping data is inspected and cleaned in Python, analytical business questions are addressed with PostgreSQL/SQL, derived features are created for segmentation, and the results are presented through an interactive dashboard.

The strongest portfolio value of the project is the combination of technical data preparation, SQL problem solving, business interpretation, and visual communication rather than any single chart or KPI.

18. Conclusion

The analysis shows a customer base of 3,900 records generating \$233,081 in recorded purchase value, with an average transaction of \$59.76 and an average rating of 3.75/5. Clothing is the leading revenue category, Young Adults contribute the largest age-group revenue, and subscribers represent 27% of the customer records without a higher average purchase value than non-subscribers.

Together, the SQL analysis, Python preparation, and dashboard provide a practical framework for understanding customer behavior and supporting merchandising, marketing, subscription, and customer-retention decisions.

19. Author

Anurag Nagarkoti

Aspiring Data Analyst | SQL | Python | Power BI | Tableau | Advanced Excel

Focused on transforming raw data into meaningful business insights through structured analysis and visualization.

Source files used for this report: customer shopping behavior CSV, Python/Pandas analysis notebook, PostgreSQL SQL analysis file, and the supplied customer behavior dashboard image.